



Faculty of Information and Technology

Assignment 01

Web Technologies

[BS CS F24 – Morning & Self Support]

Course Instructor: Dr. Noman Shafi

Instructions

- Design and develop a fully static website using only HTML, CSS, and JavaScript (no backend, server-side code, or database).
- Focus strongly on UI design ,the website should be clean, visually appealing, and well laid out.
- Focus strongly on JavaScript use it to make the website genuinely interactive, not just decorative.
- Use your own original design and content copying a ready-made template as is is not acceptable.
- Push your complete project to GitHub, using proper branches (see Version Control Requirements below).
- The marking of this assignment will be **purely based on Viva.**
- The exact submission method will be shared with you separately, closer to the deadline.

Purpose of This Assignment

This assignment is designed to familiarize you with front-end web development and User Interface (UI) design using HTML, CSS, and JavaScript. It focuses on building a real, working, static website so that you understand how structure (HTML), presentation (CSS), and behaviour (JavaScript) come together to create a usable, attractive interface.

Deadline: 14th September 2026, 12:00pm

Assignment Task

Design and develop a complete, multi-page static website using HTML, CSS, and JavaScript, on a theme/topic of your own choice (for example: personal portfolio, restaurant, educational institute, travel agency, small business, e-commerce store, event page, etc.). The website must be fully static — no backend, server-side language, database, or framework is required — and must consist of at least 4 to 5 properly interlinked pages. Strong emphasis will be placed on UI design and on meaningful use of JavaScript for interactivity.

1. Suggested Website Pages

You may use the structure below, or propose an alternate theme/page set with your instructor's approval:

1. Home Page : banner/introduction, highlights, and main navigation.
2. About Page : information about the person/business/organization.
3. Services / Products Page : listing of services or products offered.
4. Gallery / Portfolio Page : images, projects, or work samples.
5. Contact Page : contact form, location/map, and contact details.

2. Technical Requirements — HTML

- Use semantic HTML5 tags (header, nav, main, section, article, aside, footer, etc.).
- Proper, valid, and well-indented markup structure.
- Appropriate use of forms, lists, and tables wherever relevant.

3. Technical Requirements — CSS

- A single external stylesheet (or organized set of stylesheets) linked to all pages.
- Use of Flexbox and/or CSS Grid for page layout.
- A responsive design that adapts to desktop, tablet, and mobile screen sizes.
- A consistent colour scheme, typography, and spacing across all pages.

4. Technical Requirements — JavaScript

Implement at least 3–4 interactive features using JavaScript, such as:

- Client-side form validation (e.g., on the Contact page).
- An image slider/carousel.
- A responsive/hamburger navigation menu for smaller screens.
- Dynamic content such as show/hide sections, an accordion, or a modal pop-up.
- Any other meaningful DOM manipulation (adding/removing elements, changing styles on events, etc.).

5. General Requirements

- A consistent navigation bar/menu on every page, correctly linking to all other pages.
- A consistent footer across the website (e.g., contact info, copyright, social links).
- Clean folder structure : separate folders/files for HTML, CSS, JavaScript, and images.

6. Version Control Requirements

- Initialize a Git repository for your project and push it to GitHub.
- Follow a proper branching workflow :for example, create separate branches for different features/pages (e.g., feature-navbar, feature-contact-form) and merge them into your main branch as you complete them.
- Use clear, meaningful commit messages that reflect the actual progress of your work.

Marking Scheme — Viva-Based Evaluation

There are no separate written marks for this assignment. The complete evaluation will be conducted through a face-to-face Viva Voce, where you will present and explain your website live. Marks will be awarded based on:

- Overall functionality and completeness of the website (all pages working and linked correctly).
- Quality of UI design : layout, visual appeal, consistency, and attention to detail (heavily weighted).
- Meaningful and correct use of JavaScript for interactivity (heavily weighted).
- Your understanding of the HTML, CSS, and JavaScript concepts used in your own code.
- Your ability to explain, and make small live modifications to, your code when asked.
- Responsiveness of the website across different screen sizes.
- Proper use of Git/GitHub, including branching and commit history.

Viva Preparation Guidelines

Since the entire assignment is graded through viva, prepare to confidently explain and defend every part of your code. You may be asked to:

- Walk through your HTML structure and justify the semantic tags you chose.
- Explain CSS concepts used in your project : box model, selectors and specificity, Flexbox/Grid, responsive breakpoints (media queries).
- Explain your JavaScript logic : functions, events (onclick, onsubmit, addEventListener), and DOM manipulation.
- Explain your form validation logic and what happens on valid/invalid input.
- Justify your design and layout decisions from a UI/UX perspective.

Practice explaining every line of your own code before the viva , code that you cannot explain will be treated as not understood, regardless of how it looks.

Helping Materials:

<https://developer.mozilla.org/en-US/>

https://youtube.com/playlist?list=PLKyB9RYzaFRg6V_C823wgLiaySAniLKRW&si=tvSqrqD6HWM87r_A

Submission Instructions

- Push your complete, final code to your GitHub repository, with all feature branches merged into main.
- Bring your website open and running on your own laptop for the face-to-face viva.
- The exact submission method (e.g., how to share your repository link) will be communicated to you separately, closer to the deadline.